Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

Reduction of Accelerator pedal mtg bolt Half torque issue at Scorpio TCF

Mahindra and Mahindra Ltd, Nashik Plant-1, SBS.

	Plan	Actual
Start Date	**14.06.12**	**01.07.12**
Completion Date	**20.09.12**	**10.10.12**

Current Status:

92% reduction in Sept-12

(In progress)

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardise
- Step 9: Summarise & Plan future actions
- Guidelines to user

Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Shaikh.A.R	L9-O	Tool Engg	9881519976	SHAIKH.ALESHAMUDDIN@mahindra.com
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3	**Member 1**	Gokul Amrutkar	L7-O	MQA	9922408133	AMRUTKAR.GOKUL@mahindra.com
4	**Member 2**	Jayesh Jirekar	L8-O	Pro-TCF	9860961900	JIREKAR.JAYESH@mahindra.com
5	**Member 3**	Giri Debabrata	L8-O	SQA	9767744812	GIRI.DEBABRATA@mahindra.com

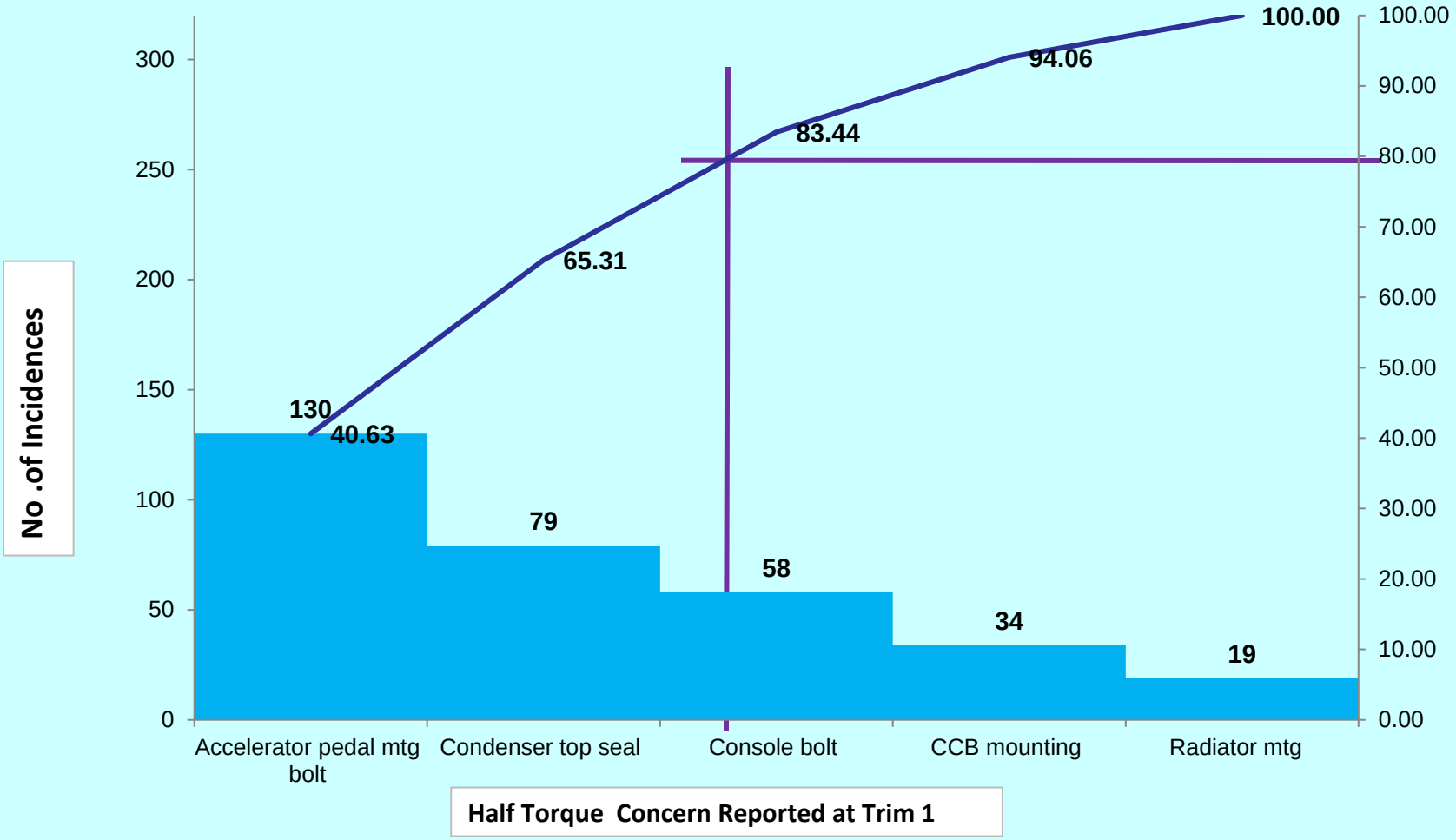
1. Select the Theme

Brief description of Problem

- What :- Accelerator pedal mtg bolt Half torque issue
- Where :- At Trim 1 in Scorpio TCF
- When :- During fitment of the accelerator pedal on bkt.
- Why :- Root cause not known
- How:- Rework qty. – 42 nos. in the month of May 2012,
  55 nos. in June-12,
  33 nos.in July -12
- Rework time – 15 to 20 min./ veh.

2. Justify the choice (reasons)

Pareto for Trim 1 Half Torque concern (3 Month data)



2. Justify the choice (reasons)

Sr No	Month	Problem Name	Location	Qty	Severity/Score	Interval (Freq)	Rework time(Cost)	Total
1	May 12 to July 12	Half torque-Accelerator pedal mtg bolt	TCF	130	9	9	9	729
2	May 12 to July 12	Half torque-Condenser top seal	TCF	79	3	9	3	81
3	May 12 to July 12	Half torque-Console bolt	TCF	58	3	9	3	81
4	May 12 to July 12	Half torque-CCB mounting	TCF	34	3	9	3	81
5	May 12 to July 12	Half torque-Radiator mtg	TCF	19	3	9	3	81

Due to High Severity, Occurrence and Rework time we have selected the "Half torque-Accelerator pedal mtg bolt" issue for resolution

2.a Immediate Containment Action

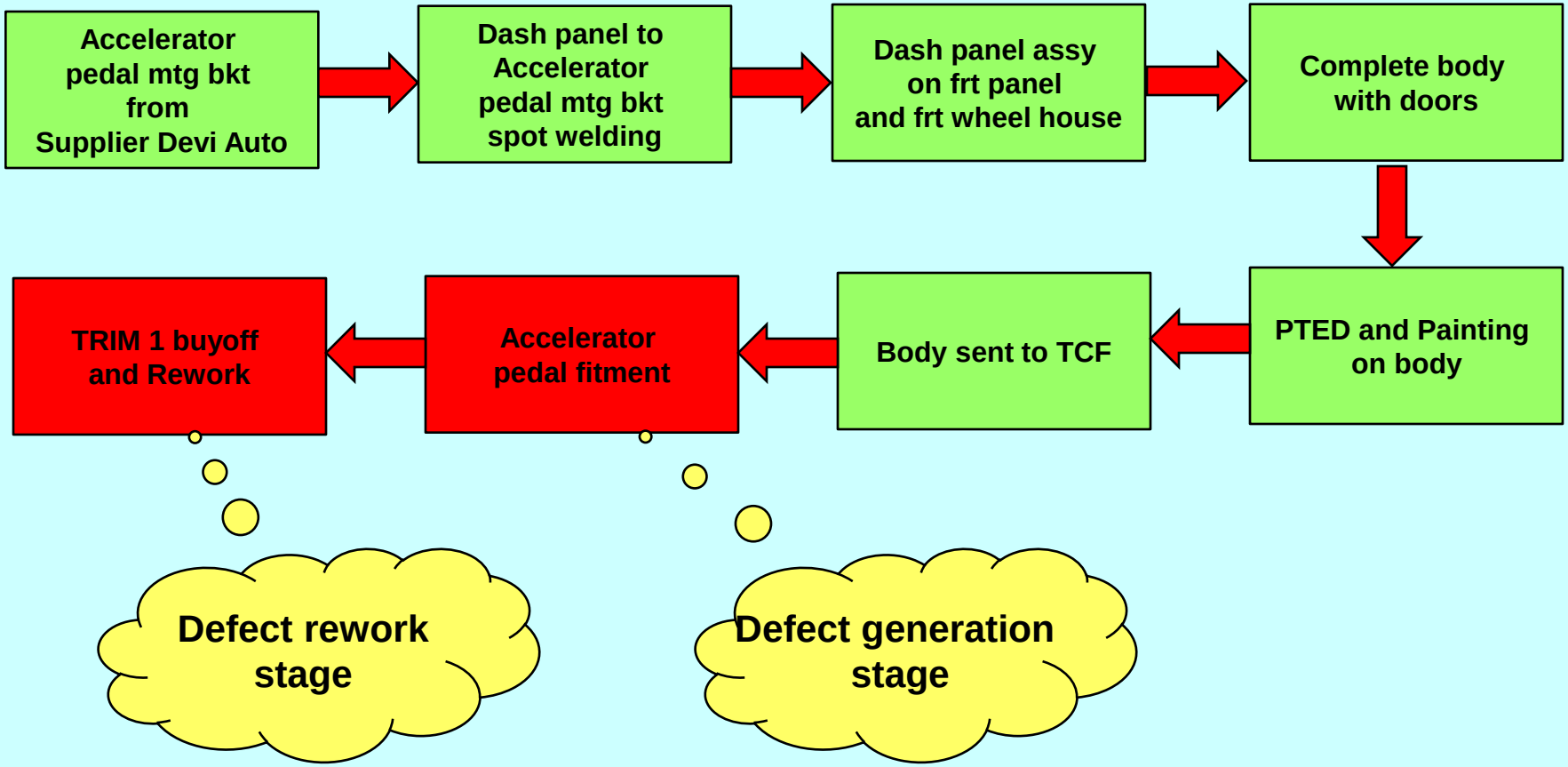
1. In order to protect the customer we have started 100% checking threading of the accelerator bkt.

2. Checkpoint added in IQ check sheet for accelerator bkt. hole mismatch

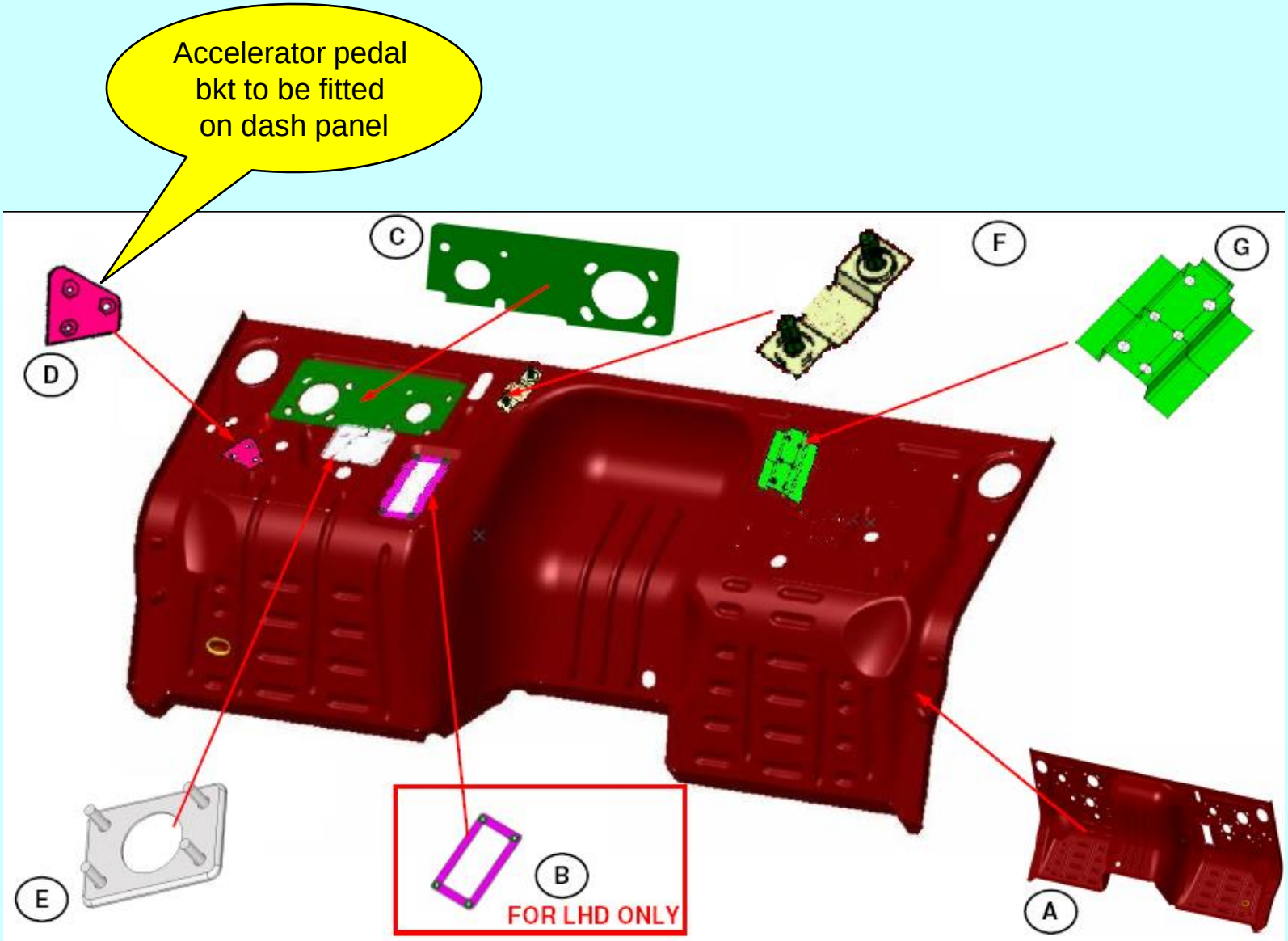
3.Rework started for defective bodies at TCF

3. Understand the current situation

Observation of process



3. Understand the current situation



3. Understand the current situation

Dash panel on fixture

Acc Pedal mtg bkt on dash assy

Total clamping

Spotting on Acc pedal mtg bkt on dash

Accelerator pedal

Accelerator Pedal Fittment in TCF

3. Understand the current situation

No clamping for Acc bkt

Acc Bkt located on 2 nut hole by pin mtg

4. Select Target

Present Level:	Avg.45 concerns Reported from May-12 To July-12 at Scorpio TCF Trim Line based on average data of last -3-months
Target:	80% Reduction in Accelerator pedal mtg. half torque concern at Trim 1 in Scorpio TCF.

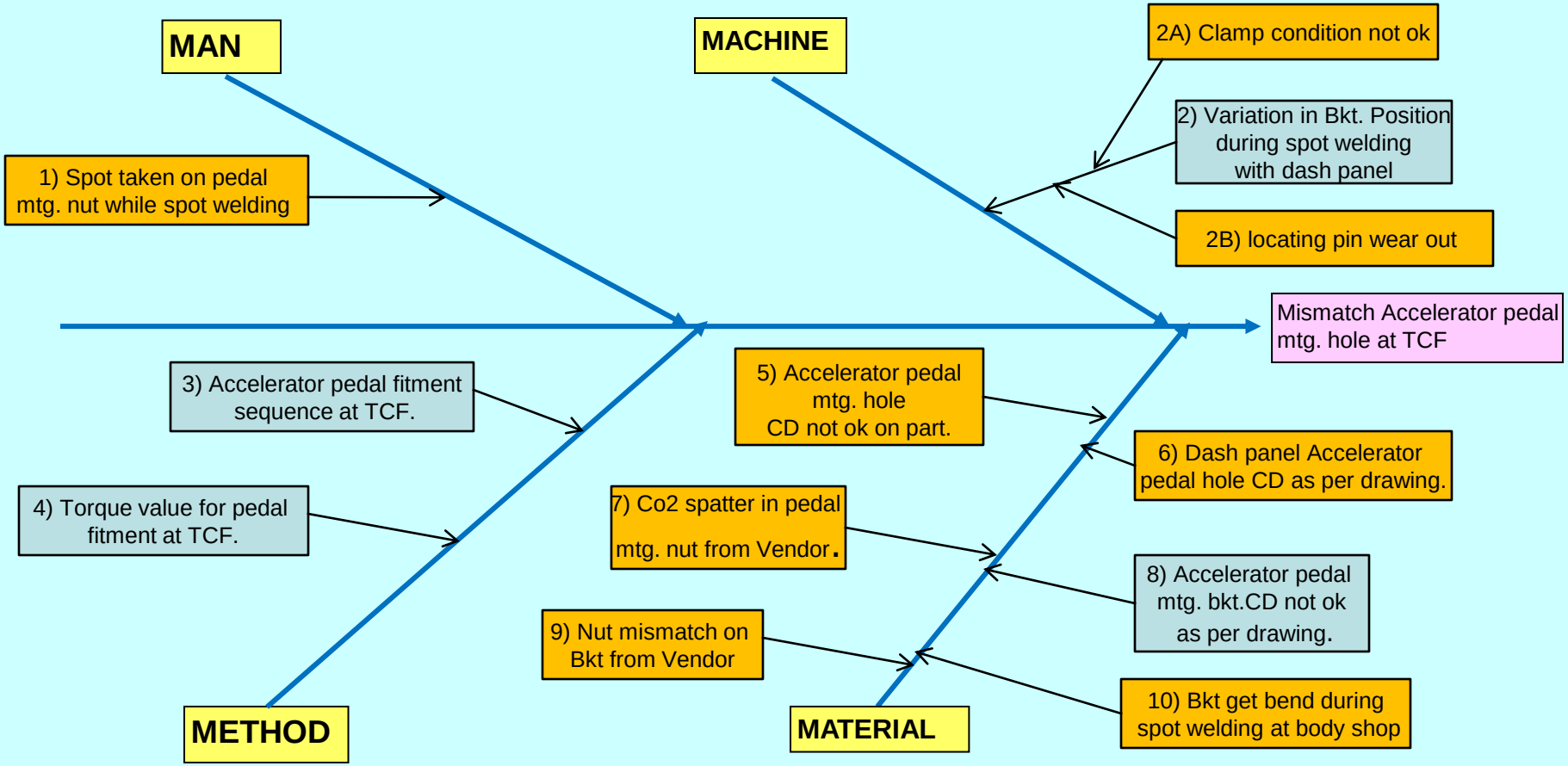
5. Analysis

Brain storming by CFT:- Possible causes at Body shop & Vendor end.

Sr. No.	Possible Causes
1)	Accelerator pedal mtg. hole CD not ok on part.
2)	Accelerator pedal mtg. bkt.CD not ok as per drawing.
3)	Variation in Bkt. Position during spot welding with dash panel due to 1) locating pin wear out 2) Clamp condition not ok.
4)	Spot taken on pedal mtg. nut while spot welding at body shop.
5)	Co2 spatter in pedal mtg. nut from Vendor.
6)	Bkt get bend during spot welding at body shop.
7)	Nut mismatch on Bkt from Vendor.
8)	Dash panel Accelerator pedal hole CD as per drawing.
9)	Accelerator pedal fitment sequence at TCF.
10)	Torque value for pedal fitment at TCF.

5. Analysis

5.1 Identify possible causes:



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr.No	Probable Cause
1	Spot taken on pedal mtg. nut while spot welding.
2A	Variation in Bkt. Position during spot welding with dash panel due to Clamp condition not ok.
2B	Variation in Bkt. Position during spot welding with dash panel due to locating pin worn out.
3	Accelerator pedal mtg.hole CD not ok on part.
4	Co2 spatter in pedal mtg. nut from Vendor.
5	Nut mismatch on Bkt from Vendor.
6	Dash panel Accelerator pedal hole CD as per drawing.
7	Bkt get bend during spot welding at body shop

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1)	Spot taken on pedal mtg. nut while spot welding.	Process walk through done, found possibility of direct spot on accelerator bkt.	**Hypothesis Valid**
2A)	Variation in Bkt. Position during spot welding with dash panel due to Clamp condition not ok.	25 dash checked found variability in bkt welding, clamp on dash panel stn.10 found loose.	**Hypothesis Valid**
2B)	Variation in Bkt. Position during spot welding with dash panel due to locating pin worn out.	Locating pin found worn out, Due to which there is excess play in nut position on Accelerator bkt.	**Hypothesis Valid**
3)	Accelerator pedal mtg.hole CD not ok on part.	CD on 5 Parts checked as per drawing, found no issue	**Hypothesis Invalid**
4)	Co2 spatter in pedal mtg. nut from Vendor.	50 Vendor parts checked on sampling basis found Co2 spatter in 7 parts	**Hypothesis Valid**
5)	Nut mismatch on Bkt from Vendor.	Sampling inspection done for vendor parts for one week, No issue found.	**Hypothesis Invalid**
6)	Dash panel Accelerator pedal hole CD as per drawing.	Accelerator pedal CD on Dash panel checked on CMM.No issue found on panel	**Hypothesis Invalid**
7)	Bkt get bend during spot welding at body shop	During spot welding at Body shop gap observed between dash & accelerator pedal bkt, due to which part get bend.	**Hypothesis Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

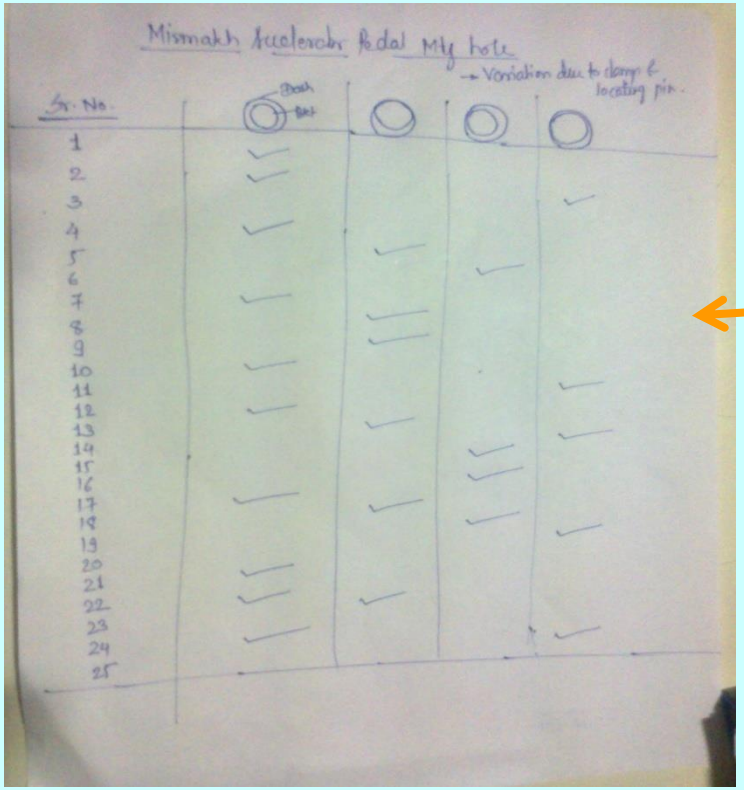
No	Probable Cause	Testing and Observations	Conclusion
1)	Spot taken on accelerator pedal mtg. nut while spot welding.	Process walk through done, found possibility of direct spot on accelerator bkt.	**Hypothesis Valid**

During Spot welding at dash panel the accelerator pedal mtg. nut get damage due to invisible spot.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
2A)	Variation in Bkt. Position during spot welding with dash panel due to Clamp condition not ok.	25 dash checked found variability in bkt welding, No clamp on dash panel stn.10.	**Hypothesis Valid**

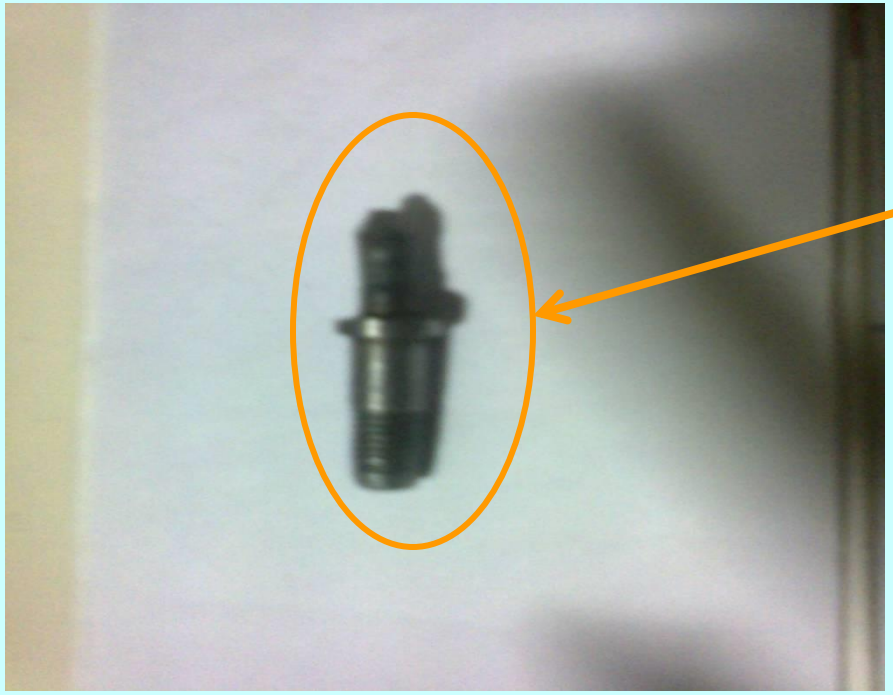


25 dash checked found variation in accelerator pedal mtg. nut.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
2B)	Variation in Bkt. Position during spot welding with dash panel due to locating pin worn out.	Locating pin found worn out, Due to which there is excess play in nut position on Accelerator bkt.	**Hypothesis Valid**



Locating pin of accelerator pedal mtg. bkt on dash panel found wear out & bend

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
4)	Co2 spatter in accelerator pedal mtg. nut from Vendor.	50 Vendor parts checked on sampling basis found Co2 spatter in 7 parts	**Hypothesis Valid**

Spatters are found in accelerator pedal mtg. nut during Co2 welding at Vendor end.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
7)	Accelerator pedal Bkt get bend during spot welding at body shop	During spot welding at Body shop gap observed between dash & accelerator pedal bkt, due to which part get bend.	**Hypothesis Valid**

Excess Gap between Dash panel & bkt during spot welding.

WHY - WHY Analysis

5.3 Identify the root causes:

Valid Probable Cause	WHY	WHY	WHY	WHY	WHY
Spot taken on accelerator pedal mtg. nut while spot welding.	Difficult gun movement	Compact area of spotting	No guiding to the gun	No provision on fixture	
Variation in Bkt. Position during spot welding with dash panel	No positive clamp for accelerator bkt positioning	No provision on fixture to clamp	Fixture design issue		

WHY - WHY Analysis

5.3 Identify the root causes:

Valid Probable Cause	WHY	WHY	WHY	WHY	WHY
Variation in Bkt. Position during spot welding with dash panel	Locating Pin undersize	Frequent wearing of the pin	Pin changing frequency not decided	Pin life not known	
Co2 spatter in accelerator pedal mtg. nut from Vendor.	Single locating pin is used for Co2 welding	No foolproof fixture available			

WHY - WHY Analysis

5.3 Identify the root causes:

Valid Probable Cause	WHY	WHY	WHY	WHY	WHY
Accelerator pedal Bkt get bend during spot welding at body shop	No positive clamp for accelerator bkt positioning	No provision on fixture	Point was not addressed during fixture design		

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Spot taken on accelerator pedal mtg. nut while spot welding.	Copper guard provided at Front End stn.10 to avoid spot on nut during spot welding.	25.07.2012	A. Shaikh	Closed

Cu Guard to avoid spot on nut.

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Variation in Bkt. Position during spot welding with dash panel	Additional Clamp provided at Dash panel stn.10 to avoid bkt movement during spot welding.	05.07.2012	A.Shaikh	Closed

Additional Clamp provided.

6. Implement the corrective measures

6.1 Actions Decided

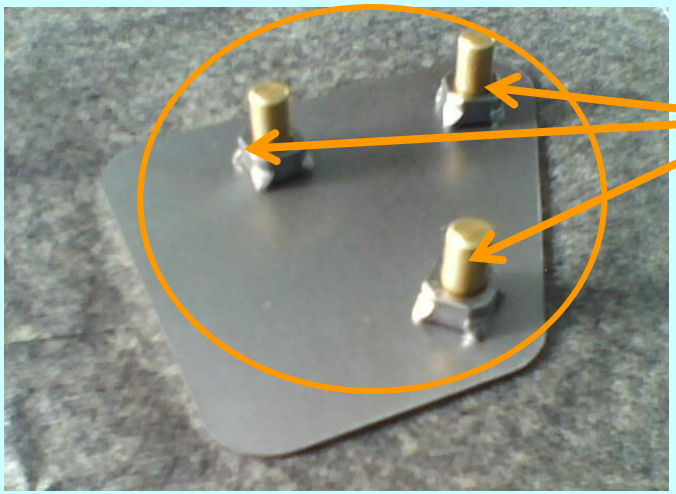
Root Cause	Action Plan	Target Dt	Resp.	Status
Variation in Bkt. Position during spot welding with dash panel	Locating pin worn out found, New Pin provided. Check point added in Supervisory check sheet.	20.07.2012	A. Shaikh	Closed

New locating Pin provided.

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Co2 spatter in accelerator pedal mtg. nut from Vendor.	ICA:- 100% Inspection & tapping at Vendor end. PCA:- 1.Providing new full proof fixture for Co2 welding 2.Co2 welding parameters are optimized	30.09.2012	D.Giri	Closed



1.Brass Pin provided for masking in nut thred avoid to spatter during co2 tack welding operation.

2.Co2 welding parameters optimized

Before	After
C-120-130Amps	C- 100-120Amps
V-22-28 V	V-20-25 V
Gas flow-10-15Kg/cm sq	Gas flow-10-15Kg/cm sq

6. Implement the corrective measures

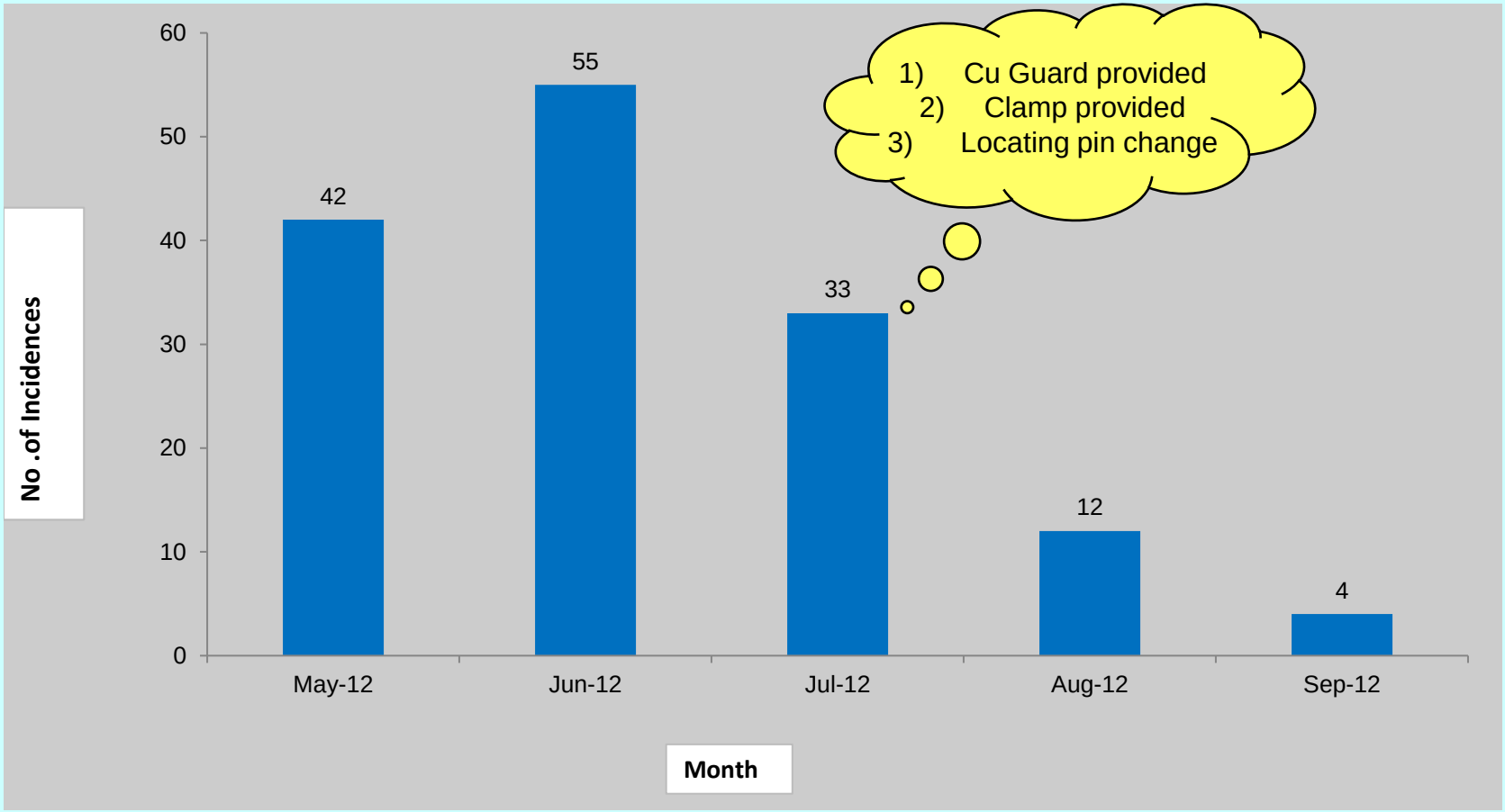
6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Accelerator pedal Bkt get bend during spot welding at body shop	Additional Clamp provided at Dash panel stn.10 to avoid bkt movement during spot welding.	05.07.2012	A.Shaikh	Closed

Additional Clamp provided to avoid the gap during spot welding.

7. Check & confirm the results & effects

7.1 Verify Results:



• 7.2 Full Scale Implementation plan:

1. All decided actions implemented.

2. Effectiveness of actions has been audited on 12.10.2012 by MQA team

Check & confirm the results & effects (Cont……)

7.3 Tangible :

1) FRC improvement at TCF by 0.8%
2) Heavy rework at TCF gets eliminated
3) High customer satisfaction through customer voice

7.3 Intangible benefits:

1. Team Morale improved
2. High Customer satisfaction
3. Team Work Built up.
4. Team Confidence Boost up.

8. Standardisation

8.1 Updation of documents:

1) Fixture History Card
2) PM Check Sheet (Locating Pin)

8.1 Updation of documents:

1) Fixture History Card

FIXTURE HISTORY CARD			FIXTURE HISTORY CARD		
ZONE:	Z03			DEPT : MFG ENGG.-SCORPIO BODY	
FIXTURE NO.	Z03 -F10		FREQUENCY : 6 Months and/or whenever changes done		
ASSY NAME:	Assy Dash panel.				
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	Remarks
10/12/05			Steering column mtg brkt wrong fitted	Stopper provided to prevent wrong fitment.	
18/10/06			Txv brkt mismatch.	adjusted by shimming.	
20/11/06			Pressure modulator brkt modification	Bock modified	En07-1592D
8/6/2007			Accelerator pedal nut mismatch. Gun inaccesible	Block modified of Accelerator pedal mounting by relieving it by 5mm	
19/8/07			Txv brkt mismatch.	Single Cu plate provided	
29/12/07			HVAC bkt mtg not possible	Stopper provided to prevent wrong fittment of connector mtg bkt	
29/02/08			Spot on edge- Clutch mtg reinf	Stopper provided to restrict the spot taken	
8/25/2008			Damage trimming hole	Pin sequence changed	
10/7/2009			Stud welding broken.(Less strength)	Stud m/c guide sleeve height increased by 10mm to 20mm	
10/7/2009			Stud welding broken.(Less strength)	Stud m/c guide sleeve height increased by 10mm to 20mm	
24/6/10			Mismatch - clutch pedal mtg hole.	Warned out locating pin replaced with new one.	QCRT L-2
14/10/10			Mismatch - clutch pedal mtg hole.	Reinforcement added to reinforce the unit.	QCRT L-2
11/2/2011			Connector mtg brkt modification.	New locating block mounted.	EN116810N
30/6/2011			Spot on hole Clutch pedal mtg brkt	stopper provided on hole to avoid spot	QCRT L-3
05.07.12			Mismatch -Acc. pedal mtg hole.	Clamp provided for gap elimination in panels	CA3P151

8.1 Updation of documents:

2) PM Check Sheet (Locating Pin)

QUALITY MANAGEMENT SYSTEM CHECK LIST

TITLE : Preventive Maintenance of Fixtures

Equipment/ Machine : Dash Assly Stn 10

Sr.No.	Jobs/Checks to be carried out	Eqt. /Matls Reqd.	Freq																	
1	Accelerator Locating Pin Dia 4.85mm	V.C.	Weekly																	
2		V.C.																		
3		V.C.																		
4		V.C.																		
5		V.C.																		
		Signature																		
		Date																		

Frequency: W-weekly, F- Fortnightly, M –Monthly, Q - Quarterly

Date	Non-conformity Observed	Disposition	Corrective Action Taken

8.2 Training:

1. OPL given to all stage operators

8.3 Additional Audit / Check

1. Supplier Audit
2. PM Check Sheet (Locating Pin)

9. Summarise & Plan future actions

9.1 Conclusion:

An innovative approach and team work by doing GEMBA study can lead to elimination of such chronic concerns

9.2 Horizontal Deployment:

Project initiated for Side Assly LH/RH Jump Seat Mounting half torque issue

9.3 Lessons Learnt:

Things Gone Wrong :

Insufficient clamping to the Accelerator pedal mtg bkt during fixture design

Things Gone Right :

We study the process implemented the additional clamp and gun guide without much disturbance to gun movement and operator fatigue

THANK YOU